

WEEK 1 – POWER BI PROJECT REPORT

Superstore Sales Data Import, Power Query & Data Cleaning

1. INTRODUCTION

Week 1 focuses on the **data preparation stage** of the Superstore Sales Analysis project. The purpose of this stage is to convert the raw sales dataset into a clean, consistent, and analysis-ready dataset using **Microsoft Power BI and Power Query Editor**.

The Superstore dataset contains information related to customers, orders, products, categories, sales, quantity, discounts, profits, and shipping. Before creating reports or visualizations, the raw data needs to be examined carefully to identify data-quality issues.

Power Query is used to perform the required **data profiling, cleaning, transformation, and validation** activities. The process includes checking column names and data types, identifying missing or incorrect values, standardizing categorical fields, verifying records, and generating additional date attributes from the Order Date column.

2. PROJECT OBJECTIVES

The main objectives of Week 1 are:

- To understand the structure and attributes of the Superstore Sales Dataset.
- To import the dataset into Microsoft Power BI.
- To use Power Query for data preparation and transformation.
- To verify the correctness of column data types.
- To identify potential data-quality problems.
- To check missing, empty, and error values.
- To verify record uniqueness and avoid unnecessary duplicate removal.
- To standardize categorical data.
- To remove unnecessary spaces and unwanted characters.
- To create meaningful date attributes from Order Date.
- To prepare a reliable dataset for further Power BI analysis.

DATASET OVERVIEW

Dataset Name:

Superstore Sales Dataset

The dataset represents sales transactions and contains information about different aspects of a retail business.

Major categories of information include:

Order Information

- Row ID
- Order ID
- Order Date
- Ship Date
- Ship Mode

Customer Information

- Customer ID
- Customer Name
- Segment

Product Information

- Category
- Sub-Category
- Product Name

Sales Information

- Sales
- Quantity
- Discount
- Profit

The dataset provides the necessary information for studying sales and profitability patterns.

DATA IMPORT PROCESS

The Superstore Sales Dataset was imported into **Power BI Desktop** using the appropriate data-import option.

After selecting the CSV file, the dataset was opened in **Power Query Editor** rather than being loaded directly. This allowed the raw data to be inspected and transformed before it was used for analysis.

The initial import process included:

1. Opening Power BI Desktop.
2. Selecting **Get Data**.
3. Selecting the CSV file.
4. Previewing the dataset.
5. Selecting **Transform Data**.
6. Opening the dataset in Power Query Editor.

Row ID	Order ID	Order Date	Ship Date	Ship Mode
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class
8	US-2023-106054	1/6/2023	1/7/2023	First Class
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class
12	US-2023-130813	1/6/2023	1/8/2023	Second Class
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class
17	US-2023-135405	1/9/2023	1/13/2023	Standard Class
18	US-2023-135405	1/9/2023	1/13/2023	Standard Class

Data Import

The **Superstore Sales CSV file** was imported into Microsoft Power BI using the **Get Data** option. After importing the dataset, the data was opened in **Power Query Editor** for detailed inspection, cleaning, and transformation. The dataset was reviewed to understand its columns, records, and overall structure before applying the required changes.

Data Cleaning & Transformation

The imported dataset was cleaned and transformed to improve its **accuracy, consistency, and usability** for further analysis. Various Power Query operations were performed to prepare the data in a structured format.

Promoted Headers

The first row of the dataset was promoted as the **column headers** so that the fields could be correctly identified and referenced during data transformation. This made the dataset easier to understand and work with in Power Query.

Data Type Correction

The data types of different columns were inspected and corrected where necessary. Appropriate data types were assigned based on the nature of each field, such as **Text, Date, Whole Number, and Decimal Number**.

One important correction was made to the **Order ID** column, which was incorrectly interpreted as a Date. It was changed to the appropriate **Text** data type to ensure that the order identifiers were displayed and handled correctly.

Cleaning Categorical Columns

The following categorical columns were cleaned and standardized:

- **Category**
- **Sub-Category**
- **Segment**

Trim and **Clean** transformations were applied to remove unnecessary spaces, unwanted characters, and formatting inconsistencies. This helped ensure that similar categorical values were treated consistently during future analysis.

Error Checking

The columns were carefully checked for:

- **Errors**
- **Empty values**
- **Incorrect data types**
- **Formatting inconsistencies**
- **Invalid or unexpected values**

The data-quality features in Power Query were used to identify potential issues. After applying the necessary corrections, the dataset was reviewed again to ensure that the transformation errors had been removed.

Perform Column Transformation

Column transformation was performed in **Power Query Editor** to organize and prepare the Superstore Sales dataset for analysis. The required columns were modified by **changing data types, renaming fields, cleaning text values, and creating new columns**. Date information from the **Order Date** column was also transformed into useful attributes such as **Year, Quarter, Month, Day, and Day of Week**. These transformations improved the structure and consistency of the dataset and made it ready for further analysis in Power BI.

Data Cleaning

Data cleaning was performed in **Power Query Editor** to improve the quality and consistency of the Superstore Sales dataset. The data was checked for **missing values, errors, incorrect data types, duplicate records, and formatting inconsistencies**. Text fields such as **Category, Sub-Category, and Segment** were cleaned using **Trim and Clean** transformations to remove unnecessary spaces and unwanted characters. After cleaning, the dataset was reviewed again to ensure that the values were consistent and suitable for further analysis in Power BI.

Table.TransformColumnTypes("#Cleaned Text1",{{"Order ID", type text}})

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-23	07-01-23	Standard Class	DP-13000	Darren Power
2	US-2023-112326	04-01-23	08-01-23	Standard Class	PO-19195	Phillina Ober
3	US-2023-112326	04-01-23	08-01-23	Standard Class	PO-19195	Phillina Ober
4	US-2023-112326	04-01-23	08-01-23	Standard Class	PO-19195	Phillina Ober
5	US-2023-141817	05-01-23	12-01-23	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
7	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
8	US-2023-106054	06-01-23	07-01-23	First Class	JO-15145	Jack O'Briant
9	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
10	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
11	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
12	US-2023-130813	06-01-23	08-01-23	Second Class	LS-17230	Lycoris Saund
13	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
14	US-2023-167199	06-01-23	10-01-23	Standard Class	ME-17320	Maria Etezad
15	US-2023-105417	07-01-23	12-01-23	Standard Class	VS-21820	Vivek Sundar
16	US-2023-105417	07-01-23	12-01-23	Standard Class	VS-21820	Vivek Sundar
17	US-2023-135405	09-01-23	13-01-23	Standard Class	MS-17830	Melanie Seite
18	US-2023-135405	09-01-23	13-01-23	Standard Class	MS-17830	Melanie Seite
19	US-2023-149020	10-01-23	15-01-23	Standard Class	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-23	15-01-23	Standard Class	AJ-10780	Anthony Jaco
21	US-2023-130092	11-01-23	14-01-23	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-23	18-01-23	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-23	16-01-23	Second Class	ND-18370	Natalie DeCh

Rename Columns

Column names were reviewed and renamed where necessary to make the dataset **clear, consistent, and easy to understand**. Meaningful names were used so that each field could be easily identified during data analysis and report development.

Table.RenameColumns("#Uppercased Text",{{"Order Date", "Order Month"}, {"Ship Date", "Ship Month"}})

Row ID	Order ID	Order Month	Ship Month	Ship Mode
1	US-2023-103800	January	January	STANDARD CLASS
2	US-2023-112326	January	January	STANDARD CLASS
3	US-2023-112326	January	January	STANDARD CLASS
4	US-2023-112326	January	January	STANDARD CLASS
5	US-2023-141817	January	January	STANDARD CLASS
6	US-2023-167199	January	January	STANDARD CLASS
7	US-2023-167199	January	January	STANDARD CLASS
8	US-2023-106054	January	January	FIRST CLASS
9	US-2023-167199	January	January	STANDARD CLASS
10	US-2023-167199	January	January	STANDARD CLASS
11	US-2023-167199	January	January	STANDARD CLASS
12	US-2023-130813	January	January	SECOND CLASS

Split Columns

The **Customer Name** column was split into two separate columns using the **Split Column by Delimiter** transformation in Power Query. The customer name was divided into **first name** and **last name** based on the space delimiter. The resulting columns, **Customer Name.1** and **Customer Name.2**, were reviewed and their data types were set to **Text**. This transformation makes the customer information more structured and allows the first and last names to be analyzed separately.

= Table.TransformColumnTypes(#"Split Column by Delimiter",{{"Customer Name.1", type text}, {"Customer Name.2", type text}}						
	APC Customer_ID	APC Customer Name.1	APC Customer Name.2	APC Segment	APC Country/Region	
	Valid 100%	Valid 100%	Valid 99%	Valid 100%	Valid 100%	
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	
	Empty 0%	Empty 0%	Empty < 1%	Empty 0%	Empty 0%	
1	DP-13000	Darren	Powers	Consumer	United States	
2	PO-19195	Phillina	Ober	Home Office	United States	
3	PO-19195	Phillina	Ober	Home Office	United States	
4	PO-19195	Phillina	Ober	Home Office	United States	
5	MB-18085	Mick	Brown	Consumer	United States	
6	ME-17320	Maria	Etezadi	Home Office	United States	
7	ME-17320	Maria	Etezadi	Home Office	United States	
8	JO-15145	Jack	O'Briant	Corporate	United States	
9	ME-17320	Maria	Etezadi	Home Office	United States	
10	ME-17320	Maria	Etezadi	Home Office	United States	
11	ME-17320	Maria	Etezadi	Home Office	United States	
12	LS-17230	Lycoris	Saunders	Consumer	United States	
13	ME-17320	Maria	Etezadi	Home Office	United States	
14	ME-17320	Maria	Etezadi	Home Office	United States	
15	VS-21820	Vivek	Sundaresam	Consumer	United States	
16	VS-21820	Vivek	Sundaresam	Consumer	United States	
17	MS-17830	Melanie	Seite	Consumer	United States	

Trimmed Text

The **Trim** transformation was applied in Power Query to remove unnecessary **leading and trailing spaces** from text values. This was useful for maintaining consistency in columns containing customer and categorical information. By removing extra spaces, values that should be identical are stored in a uniform format, which improves the accuracy of **filtering, grouping, sorting, and analysis** in Power BI.

= Table.TransformColumns(#"Changed Type1",{{"Customer Name", Text.Trim, type text}})						
	Order Date	Ship Date	APC Ship Mode	APC Customer ID	APC Customer Name	
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	
1	1/3/2023	1/7/2023 12:00:00 AM	STANDARD CLASS	DP-13000	Darren Powers	
2	1/4/2023	1/8/2023 12:00:00 AM	STANDARD CLASS	PO-19195	Phillina Ober	
3	1/4/2023	1/8/2023 12:00:00 AM	STANDARD CLASS	PO-19195	Phillina Ober	
4	1/4/2023	1/8/2023 12:00:00 AM	STANDARD CLASS	PO-19195	Phillina Ober	
5	1/5/2023	1/12/2023 12:00:00 AM	STANDARD CLASS	MB-18085	Mick Brown	
6	1/6/2023	1/10/2023 12:00:00 AM	STANDARD CLASS	ME-17320	Maria Etezadi	
7	1/6/2023	1/10/2023 12:00:00 AM	STANDARD CLASS	ME-17320	Maria Etezadi	
8	1/6/2023	1/7/2023 12:00:00 AM	FIRST CLASS	JO-15145	Jack O'Briant	
9	1/6/2023	1/10/2023 12:00:00 AM	STANDARD CLASS	ME-17320	Maria Etezadi	
10	1/6/2023	1/10/2023 12:00:00 AM	STANDARD CLASS	ME-17320	Maria Etezadi	
11	1/6/2023	1/10/2023 12:00:00 AM	STANDARD CLASS	ME-17320	Maria Etezadi	
12	1/6/2023	1/8/2023 12:00:00 AM	SECOND CLASS	LS-17230	Lycoris Saunders	

Date & Time Transformation

The **Order Date** column was transformed in Power Query to create additional date-related fields for better analysis of sales trends. The following attributes were extracted from the Order Date:

- **Year** – identifies the year in which the order was placed.
- **Quarter** – divides the year into four quarters.
- **Month** – identifies the month of the order.
- **Day** – identifies the day of the month.
- **Day of Week** – identifies the weekday on which the order was placed.

These transformations help organize the sales data based on different time periods and make it easier to analyze **monthly, quarterly, yearly, and weekly sales patterns**.

= Table.TransformColumns(#"Extracted Month Name", {"Order Date", each Date.DayOfWeekName(,), type								Query Settings	
Order ID		Order Date		Ship Date		Ship Mode		Customer ID	
Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%		Valid 100% Error 0% Empty 0%	
1	US-2023-103800	Tuesday	January	STANDARD CLASS	DP-13000				
2	US-2023-112326	Wednesday	January	STANDARD CLASS	PO-19195				
3	US-2023-112326	Wednesday	January	STANDARD CLASS	PO-19195				
4	US-2023-112326	Wednesday	January	STANDARD CLASS	PO-19195				
5	US-2023-141817	Thursday	January	STANDARD CLASS	MB-18085				
6	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
7	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
8	US-2023-106054	Friday	January	FIRST CLASS	JO-15145				
9	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
10	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
11	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
12	US-2023-130813	Friday	January	SECOND CLASS	LS-17230				
13	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
14	US-2023-167199	Friday	January	STANDARD CLASS	ME-17320				
15	US-2023-105417	Saturday	January	STANDARD CLASS	VS-21820				
16	US-2023-105417	Saturday	January	STANDARD CLASS	VS-21820				

Power Query – Applied Steps

The **Applied Steps** section in Power Query records all the data cleaning and transformation operations performed on the Superstore Sales dataset. Each step represents a specific modification made to the data and is automatically saved in the order in which it was applied.

The main applied steps used in this project include:

- **Source** – Imported the Superstore Sales CSV dataset.
- **Promoted Headers** – Converted the first row into column headers.
- **Changed Type** – Assigned appropriate data types to the columns.
- **Uppercased Text** – Standardized selected text values.
- **Trimmed Text** – Removed unnecessary spaces from text fields.
- **Extracted Month Name** – Created the month name from the date field.
- **Extracted Day Name** – Created the day name from the date field.

- **Renamed Columns** – Changed column names to make them clear and meaningful.
- **Split Column by Delimiter** – Divided the Customer Name column into separate fields.
- **Changed Type2** – Applied the final required data types to the transformed columns.

The **Applied Steps** feature provides a clear record of the entire transformation process and makes it easier to review, modify, or reproduce the data preparation process.

3. Tasks Performed

Task 1: Import Superstore Dataset

The Superstore Sales Dataset was imported into **Power BI Desktop** and opened in **Power Query Editor** for data preparation.

Task 2: Inspect Dataset

The available columns and records were examined to understand the structure and contents of the dataset.

Task 3: Verify Data Types

The data types of the columns were checked and corrected where necessary. Date, text, whole number, and decimal number fields were assigned appropriate data types.

Task 4: Check Data Quality

The dataset was checked for **missing values, errors, and duplicate records** using Power Query's data-quality features.

Task 5: Clean and Standardize Text

The **Category, Sub-Category, and Segment** columns were cleaned using **Trim** and **Clean** transformations to remove unnecessary spaces and unwanted characters.

Task 6: Transform Columns

Column names were reviewed and renamed where required to maintain a clear and consistent structure.

Task 7: Extract Date Attributes

Additional columns were created from **Order Date**:

- Year
- Quarter

- Month
- Day
- Day of Week

Task 8: Apply Transformations

After completing the cleaning and transformation process, **Close & Apply** was used to load the prepared dataset into Power BI.

4. Deliverables

The Week 1 project deliverables are:

- Imported **Superstore Sales Dataset** into Power BI.
- Cleaned and transformed the dataset using **Power Query Editor**.
- Verified appropriate **data types**.
- Checked **missing values, errors, and duplicate records**.
- Standardized **Category, Sub-Category, and Segment**.
- Created **Year, Quarter, Month, Day, and Day of Week** columns.
- Loaded the cleaned dataset into Power BI using **Close & Apply**.
- Saved the completed **Power BI (.pbix) project file**.

5. Conclusion

Week 1 successfully completed the **data import, cleaning, and transformation** stage of the Superstore Sales project. The raw dataset was prepared using Power Query and converted into a clean and structured dataset suitable for analysis in Power BI. The completed data preparation provides a reliable foundation for the project's further analysis.